

BO preliminaries

- Introduction to BO
- Basic terminology

Optimization problems

An optimization problem is denoted with

$$\min_{x \in \mathcal{X}} f(x) \quad \text{or} \quad \max_{x \in \mathcal{X}} f(x)$$

The **objective** is a function $f : \mathcal{X} \mapsto \mathbb{R}$

The **search space** is a set $\mathcal{X} \subset \mathbb{R}^d$

The **queries** are points $x \in \mathcal{X}$

Optimization types are trivially inter-convertible

$$\min_{x \in \mathcal{X}} f(x) = \max_{x \in \mathcal{X}} g(x)$$

$$g(x) = -f(x)$$

$$\max_{x \in \mathcal{X}} f(x) = \min_{x \in \mathcal{X}} g(x)$$

$$g(x) = -f(x)$$

Similar conversions are achieved by any strictly decreasing function $h : \mathbb{R} \mapsto \mathbb{R}$

$$g = h \circ f$$

Paradigms of optimization

Computational schemes for the solution of optimization problems (generally) follow one of two paradigms

- In numerical optimization (e.g. Newton-Raphson, Nelder-Mead, etc)
the objective is relatively **cheap** to compute
- In stochastic optimization (e.g. Bayesian optimization, stochastic gradient descent, etc)
the objective is relatively **expensive** to compute

Algorithms in both paradigms proceed **iteratively**

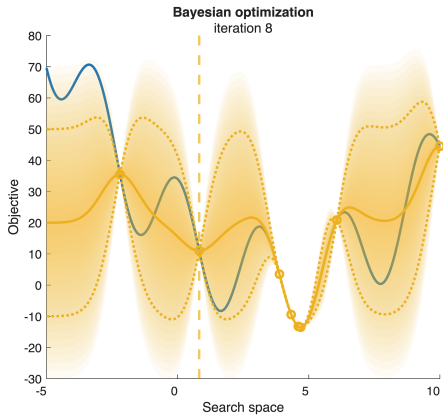
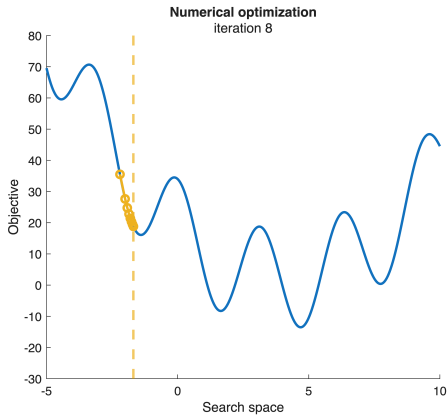
- start with one or more preselected query points
- repeat as needed:
 - compute the objective values at the queries
 - decide on new queries

Within the main loop

- numerical algorithms select queries **directly** via the objective
- stochastic algorithms select queries **indirectly** via surrogates/emulators of the objective

Bayesian optimization loops are characterized by the sequential development and deployment of global **probabilistic** surrogates

Demonstration of numerical and Bayesian optimization procedures



Characteristics of BO

Bayesian optimization loops

- rely on probabilistic surrogates of the objective
- develop and refine each surrogate sequentially
- select queries via the surrogates

...apply Gaussian processes

...apply Bayesian updating

...apply decision theory

A **Gaussian process** is a non-parametric probabilistic model used to model random functions

...normal/Gaussian random variables are used for uncertain scalars

...multivariate normal/Gaussian random variables are used for uncertain vectors

...Gaussian processes are used for uncertain functions

Bayesian updating is a mathematical method for modifying the probabilities of an uncertain quantity of interest as new data become available

...Bayes' rule

Decision theory is the study and application of mathematical and logical frameworks to analyze how choices are made under uncertainty

...optimal actions such as new query selection and termination criteria

The main BO loop

The main algorithmic loop in a BO scheme

- updates data summaries
- updates the surrogate
- obtains a new acquisition function
- maximizes the acquisition function
- obtains new queries
- evaluates the objective at new queries

...expand existing data structures

...refine existing Gaussian process

...byproduct of Gaussian process

...numerical optimization

...select actions

In our context, **data** consist of already selected queries and past objective evaluations

A **surrogate** is a probabilistic model built on data to approximate the objective and provide **both** a prediction and a measure of its uncertainty at every query in the search space

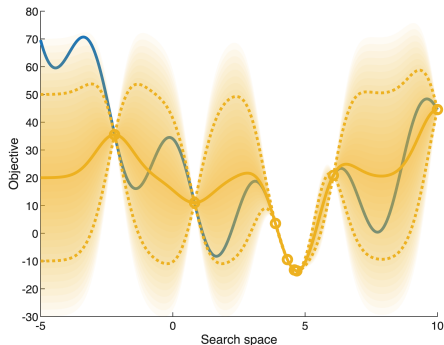
An **acquisition function** is a mathematical device that uses the surrogate's predictions and uncertainty to rank every query according to its utility for future evaluation

An **optimization policy** is the set of rules that transform data to new query selections

Exploration vs exploitation

The exploration vs exploitation dilemma is the core challenge in selecting queries

- **Exploitation:** Sampling in regions of the search space where the surrogate indicates the objective yields global **optima**
...the aim is to locate optimizers more accurately
- **Exploration:** Sampling in regions of the search space where the surrogate indicates high **uncertainty**
...the aim is to identify new optimizers

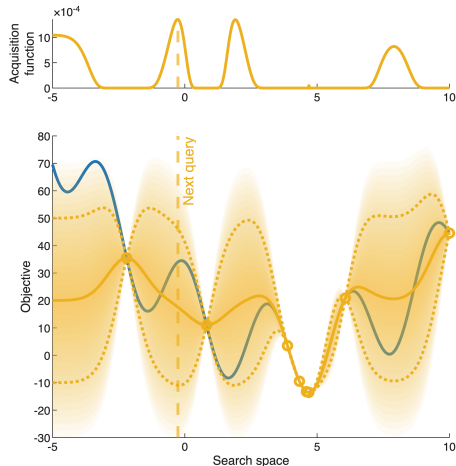


Acquisition functions

An acquisition function takes the predictions and uncertainty generated by the current surrogate and converts them into a numerical score for **every** query in the search space

The **score** considers the selection of subsequent queries as a rational decision under uncertainty and balances two competing drives

- What the model expects ...this is exploitation
against
- How uncertain the model is ...this is exploration



BO roadmap

Given an optimization problem

$$\min_{x \in \mathcal{X}} f(x) \quad \text{or} \quad \max_{x \in \mathcal{X}} f(x)$$

Bayesian optimization reduces to

- a choice of the **surrogate** ...Gaussian processes... ...kernel configuration and adaptation
- a description of the **data** ...multidimensional arrays... ...high performance computing
- a rigorous derivation of the **policy** ...Bayesian decision theory... ...utility representations